Nome _____ N.º Aluno _____

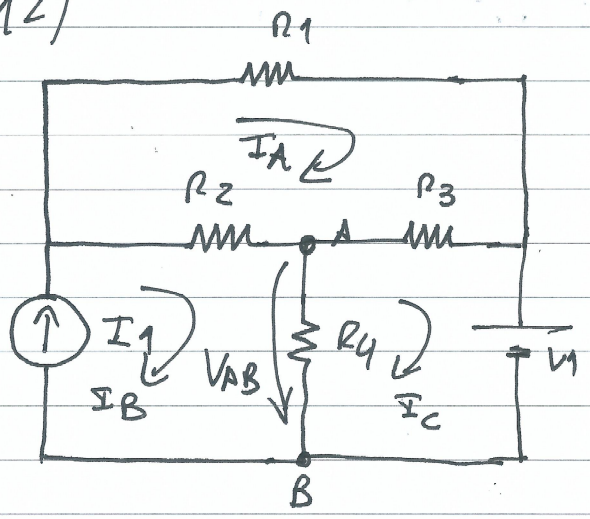
Curso AULA 5 Ano Letivo ___/___ Data da Avaliação ___/___/___

Prova Escrita de: _____ N.º Folhas _____

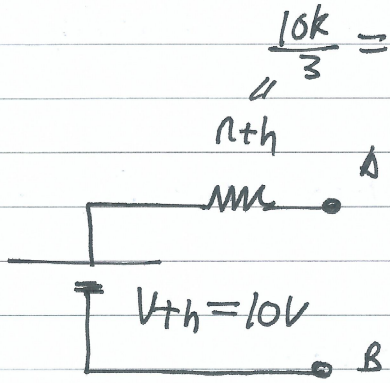
Época: _____



12)




$\Rightarrow$




a) CALCULO $V_{th}$ = ?

$$\begin{cases} \bullet\ R_1 I_A + R_3 (I_A - I_C) + R_2 \cdot (I_A - I_B) = 0 \\ \bullet\ R_3 (I_C - I_A) + V_1 + R_4 \cdot (I_C - I_B) = 0 \\ \bullet\ I_B = I_1 \end{cases} \Rightarrow I_A, I_B, I_C$$

$$V_{AB} = R_4 \times (I_B - I_C)$$

$$I_A \cdot (R_1 + R_3 + R_2) + I_B \cdot (-R_2) + I_C \cdot (-R_3) = 0$$

$$I_A \cdot (-R_3) + I_B \cdot (-R_4) + I_C \cdot (R_3 + R_4) = -V_1 \Rightarrow$$

$$I_B = I_1$$

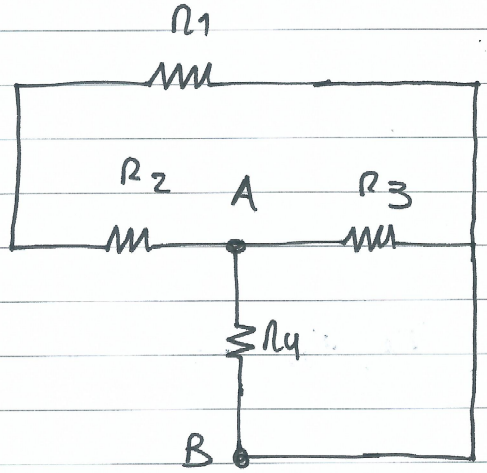
$$\begin{cases} I_A \cdot (R_1 + R_3 + R_2) + I_C \cdot (-R_3) = I_B \cdot R_2 \\ I_A (-R_3) + I_C (R_3 + R_4) = I_B \times R_4 - V_1 \end{cases}$$

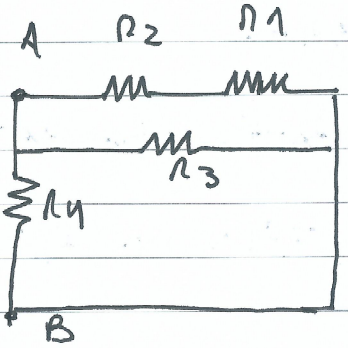
$$I_B = I_1$$

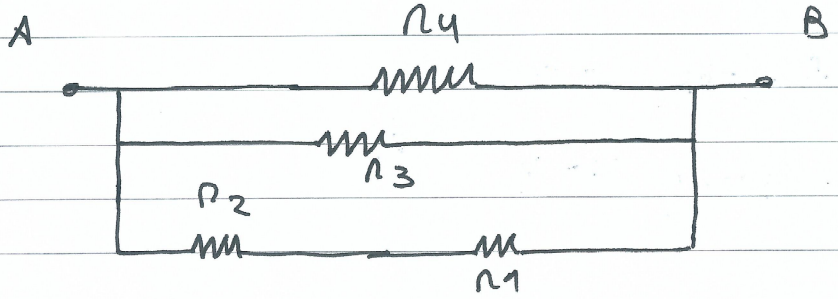
$$\begin{bmatrix} (R_1+R_2+R_3) & -R_3 \\ -R_3 & (R_3+R_4) \end{bmatrix} \begin{bmatrix} I_A \\ I_C \end{bmatrix} = \begin{bmatrix} I_B \times R_2 \\ I_B \times R_4 - V_1 \end{bmatrix}$$

$$V_{AB} = R_4 \times (I_B - I_C) = 10V$$

CALCULATION =.?




$$R_{th} = R_4 // R_3 // (R_2+R_1) = 10k // 10k // (4k+6k)$$

$$= \frac{10k}{3}$$

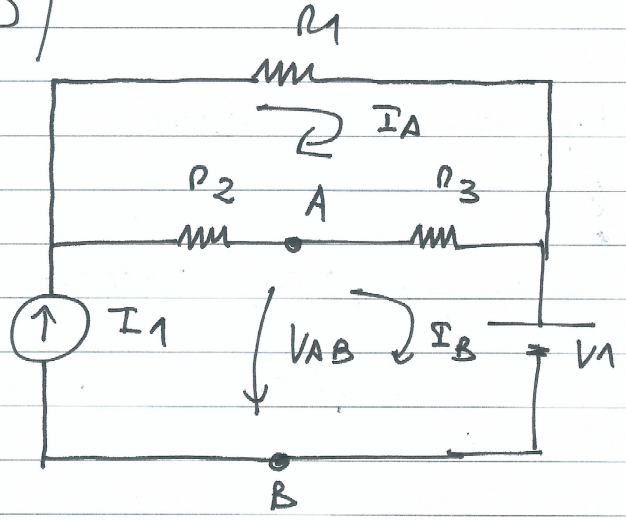
Nome _____ N.º Aluno _____

Curso _____ Ano Letivo ___ / ___ Data da Avaliação ___ / ___ / ___

Prova Escrita de: _____ N.º Folhas _____

Época: _____ (3)

b)




$V_{th} = V_{AB} = 15V$

$$\begin{cases} R_1 \cdot I_A + R_3 \cdot (I_A - I_B) + R_2 \cdot (I_A - I_B) = 0 \\ I_B = I_1 \end{cases}$$

$$(R_1 + R_3 + R_2)\, I_A = I_B \cdot (R_3 + R_2)$$

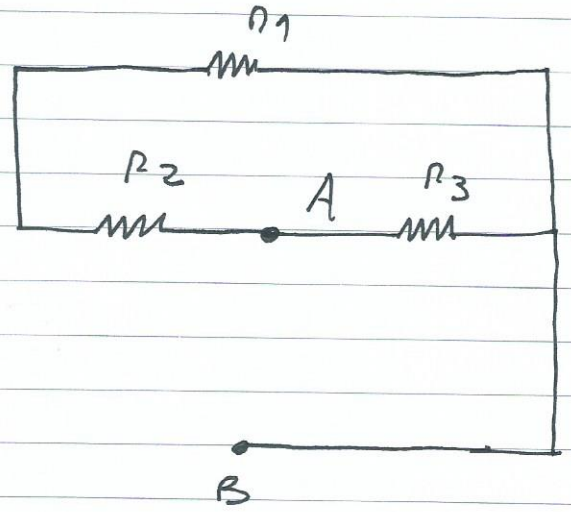
$I_B = I_1$ $\qquad I_B = 5mA$

$$I_A = \frac{I_1 \times (R_3 + R_2)}{R_1 + R_2 + R_3} = \frac{5m \times 16K}{20K} = \frac{15K\Omega}{4} = 4\,mA$$

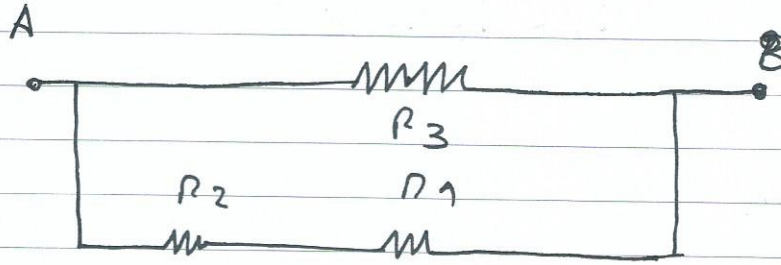
$$V_{AB} = R_3 \cdot (I_B - I_A) + V_1$$

$$V_{AB} = 10K \cdot (5m - 4m) + 5 = 10 + 5 = 15V$$

(4)

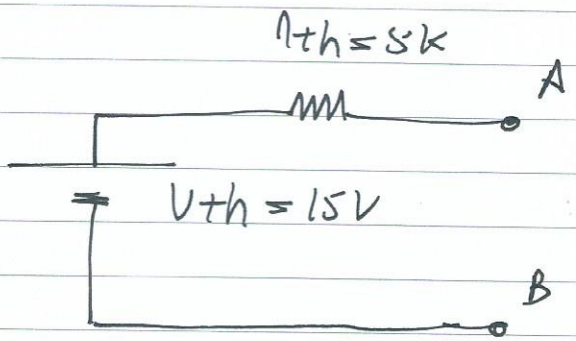



$R_{th} = R_3 // (R_2 + R_1)$

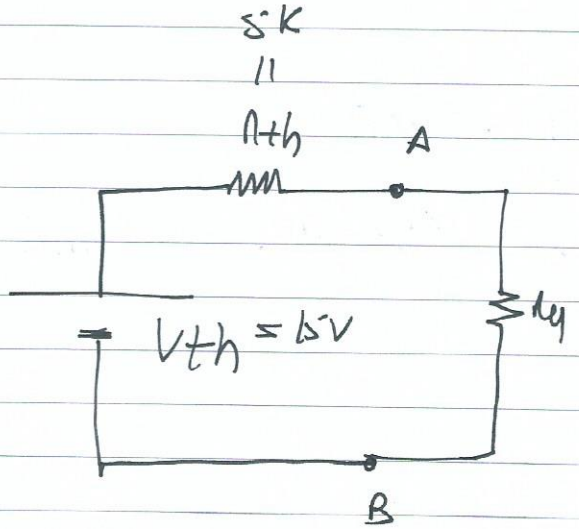



$R_{th} = 10k // (5k + 4k)$

$= 5k$




c)

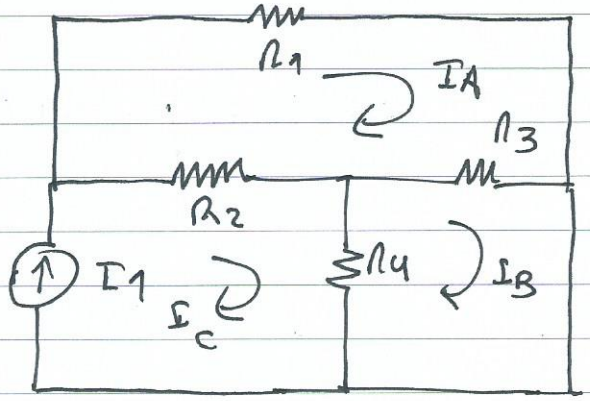



$$I_{RL} = \frac{V_{th}}{R_{th} + R_L}$$

$$I_{RL} = \frac{15}{5k + 10k} = 1\,mA$$

d) contributo $I_1$ = ?

($\downarrow$) $I_{R4}'$ = ?




$$I_{R4}' = I_C - I_B$$

$$(\downarrow)\ I_{R4}' = I_1 - I_B = 9(6)\,mA$$

- $I_C = I_1$
- $R_1 \cdot I_A + R_3 \cdot (I_A - I_B) + R_2 \cdot (I_A - I_C) = 0$
- $R_3 \cdot (I_B - I_A) + R_4 \cdot (I_B - I_C) = 0$

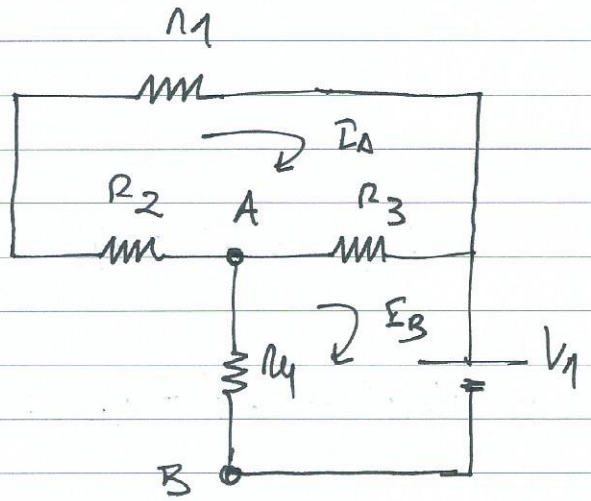
$$I_A \cdot (R_1 + R_3 + R_2) + I_B \cdot (-R_3) = R_2 \cdot I_C$$

$$I_A \cdot (-R_3) + I_B \cdot (R_3 + R_4) = R_4 \cdot I_C$$

$I_C = I_1$

$$\begin{bmatrix} (R_1+R_2+R_3) & (-R_3) \\ -R_3 & (R_3+R_4) \end{bmatrix} \begin{bmatrix} I_A \\ I_B \end{bmatrix} = \begin{bmatrix} R_2 \cdot I_C \\ R_4 \cdot I_C \end{bmatrix}$$

c) contributo $V_1$.




$$(\downarrow) I_{R4}'' = -I_B$$

$$I_{R4}'' = 9(3)\,mA$$

$$(R_1+R_2)\cdot I_A + R_3\cdot(I_A - I_B) = 0$$

$$V_1 + R_4\cdot I_B + R_3\,(I_B - I_A) = 0$$

$$I_A\cdot(R_1+R_2+R_3) + I_B\cdot(-R_3) = 0$$

$$I_A\cdot(-R_3) + I_B\cdot(R_4+R_3) = -V_1$$

$$\Rightarrow \begin{bmatrix} (R_1+R_2+R_3) & -R_3 \\ -R_3 & (R_4+R_3) \end{bmatrix} \begin{bmatrix} I_A \\ I_B \end{bmatrix} = \begin{bmatrix} 0 \\ -V_1 \end{bmatrix}$$

d) $I_{R4}(\downarrow) = \underset{(\downarrow)}{I_{R4}'} + \underset{(\downarrow)}{I_{R4}''} = 9(6)\,m + 9(3)\,m$

$= 1\,mA$

Nome _____ N.º Aluno _____

Curso _____ Ano Letivo ____/____ Data da Avaliação ____/____/____

Prova Escrita de: _____ N.º Folhas _____

Época: _____

(7)

g)

$P_{R_4} = R_4 \cdot I_{R_4}^2 = 10k \times (1m)^2$

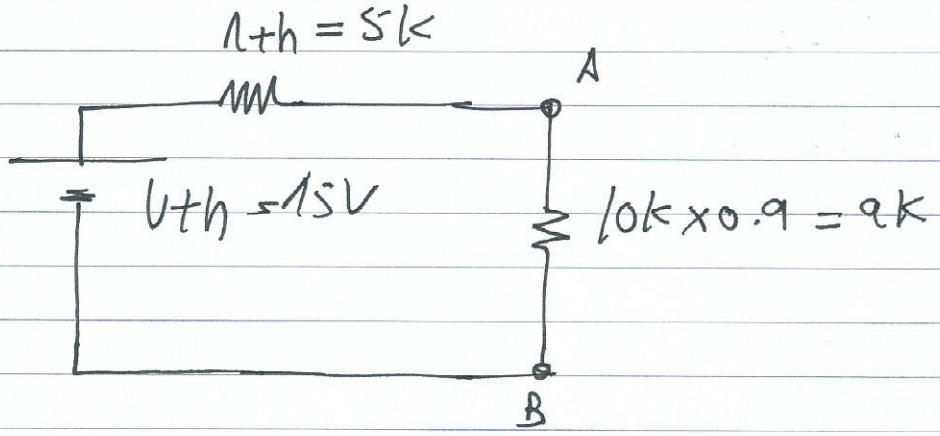
$= 10k \times 10^{-6} = 10^4 \times 10^{-6} = 0,01\,W$

$= \underline{10\,mW}$

$P = \frac{E}{T} \Rightarrow E = P \cdot T = 0,01\,(J/s) \times 10 \times 60 \times 60\,s$

$E_J = 0,01 \times 10 \times 3600\,J = 360\,J$

h) $R_4(10\%) = 9 \times R_4 = \underline{9k}$




$I_{R_4} = \frac{15}{5k+9k}$

$I_{R_4} \cong 1,07143\,mA$